ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) | OMR, Chennai - 600119

PROFESSIONAL ENGLISH LAB INTEGRATED

STUDENT PORTFOLIO

A learning record in communication, personal development and Physical AI engineering

STUDENT DETAILS	
Student Name	S Gokul
Roll Number	26CS152
Department	B.E. Computer Science and Engineering
Class / Section	CSE - C
Academic Batch	2026 - 2030
Subject	Professional English Lab Integrated
Class Code	EN25C09

DEPARTMENT OF HUMANITIES AND SCIENCES

Affiliated to Anna University | Chennai, Tamil Nadu

COLLEGE PORTFOLIO | 2026

S GOKUL

COMPUTER SCIENCE & ENGINEERING

First-Year B.E. Student

Aspiring Physical AI Engineer | Robotics and Intelligent Systems

EDUCATION

B.E. Computer Science and Engineering
2026 - 2030
St. Joseph's College of Engineering, Chennai

TECHNICAL FOUNDATION

C • C++ • Python • Java • SQL • Linux

CAREER DIRECTION

Physical AI • Robotics • Computer Vision • Intelligent Systems

"My goal is to turn intelligent algorithms into useful systems that can sense, reason and act in the real world."

Learning with curiosity. Building with purpose. Improving through practice.

PORTFOLIO GUIDE

Contents and Page Guide

Professional English Lab Integrated | Student portfolio | 2026

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Editorial note: This portfolio is an original student document prepared from S Gokul's supplied resume, self-introduction, career roadmap, project proposal, English level records, NPTEL learning report and biographical note. The sample portfolio is used only as a structural and visual reference. Wording, layout and colour system have been redesigned.

01 | PERSONAL PROFILE

About Me, Education and Technical Profile

An overview of my academic foundation, interests and developing capabilities.

Profile

I am S Gokul, a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai. I am interested in the intersection of artificial intelligence and the physical world, particularly robotics, computer vision and intelligent systems. I want to build a strong foundation in computing before progressing towards practical AI-enabled machines.

Profile detail	Information
Full name	S Gokul
Institution	St. Joseph's College of Engineering, Chennai
Degree and department	B.E. Computer Science and Engineering
Year / batch	First Year | 2026–2030
Class / section	CSE - C
Roll number	26CS152
Career aspiration	Physical AI Engineer / Robotics professional

Technical foundation

Area	Current learning / knowledge
Programming	C, C++, Python and Java
Database	SQL fundamentals
Computer science	Programming fundamentals, logical thinking and problem-solving
Development tools	Linux, Git, GitHub and Visual Studio Code
AI and robotics interests	Machine learning, computer vision, robotics and autonomous systems

Learning credentials

- Object-Oriented Programming Concepts in C++ — Great Learning.
- C for Beginners — Great Learning.

02 | CAREER DIRECTION

My Path Towards Physical AI Engineering

A progressive plan that connects software foundations with intelligent physical systems.

Career objective

My long-term objective is to become a Physical AI Engineer capable of developing systems that perceive their surroundings, interpret information, make decisions and perform useful physical actions. I plan to combine programming, artificial intelligence, computer vision, robotics, sensors, embedded computing and control systems through continuous study and project work.

Stage	Learning focus	Target outcome
Year 1 **2026–27**	Python and C++; mathematics; Linux; electronics basics; sensors; Arduino or Raspberry Pi; introductory computer vision.	A sensor-based mini project and a computer-vision prototype, documented clearly.
Year 2 **2027–28**	ROS 2; robot kinematics; perception; OpenCV; deep learning; sensor fusion; Gazebo or Isaac Sim.	A simulated autonomous robot integrating camera or sensor data.
Year 3 **2028–29**	Reinforcement learning; SLAM; advanced perception; manipulation; control systems; embedded and edge AI.	An intelligent robot demonstrating perception, navigation or manipulation.
Year 4 **2029–30**	Physical AI architecture; multimodal models; task planning; human-robot interaction; optimization and deployment.	An integrated Physical AI capstone with a strong technical portfolio.

Development principles

- **Understand before implementing:** plan the architecture, data flow and control logic before coding.
- **Learn through building:** convert each major topic into a small experiment, simulation or working prototype.
- **Document the process:** maintain readable code, GitHub repositories, diagrams, test results and project notes.
- **Test responsibly:** validate sensors, control limits and safety behaviour before operating physical hardware.

My guiding idea

> ***"I do not want to study AI only as software. I want to understand how intelligent systems can interact safely and effectively with the physical world."***

03 | SELF-ASSESSMENT

Personal SWOC Analysis

A realistic review of internal strengths and areas for development, alongside external opportunities and challenges.

STRENGTHS

- Curiosity about AI, robotics and emerging technologies.
- Interest in programming and logical problem-solving.
- Willingness to learn new tools and technical concepts.
- Motivation to connect software with real-world applications.

WEAKNESSES

- Robotics, control theory and embedded systems are still developing areas.
- Limited hands-on experience with hardware, sensors and actuators.
- Need to improve system planning before beginning implementation.
- More practice is required in advanced algorithms and technical integration.

OPPORTUNITIES

- Growing interest in Physical AI, autonomous machines and robotics.
- Open-source tools such as ROS 2, OpenCV and simulation platforms.
- Access to online courses, research communities and project repositories.
- Opportunities to gain experience through competitions and internships.

CHALLENGES

- Physical AI requires knowledge across software, hardware, mathematics and control.
- Tools and frameworks evolve quickly.
- Real-world systems can be difficult to debug due to sensors, latency and environmental conditions.
- Entry-level roles require demonstrable practical ability.

Action plan

- Strengthen Python, C++, mathematics and data structures through regular practice.
- Use simulation tools before moving complex experiments to physical hardware.
- Learn sensors, microcontrollers, actuators and basic electronics through small builds.
- Create a project plan with architecture, milestones, test cases and safety checks before implementation.
- Publish well-documented projects and seek feedback from mentors and technical communities.

04 | FEATURED PROJECT

AI-Powered Smart Campus Energy Management System

A proposed Physical AI application combining computer vision, machine learning, IoT and controlled actuation.

Project overview

The project aims to reduce unnecessary electricity consumption in classrooms and laboratories by using AI to understand room occupancy. A camera provides visual input, a computer-vision model estimates the number of people present, and an intelligent decision layer determines whether selected electrical loads should remain active. An edge controller communicates with an isolated relay interface to carry out the approved action.

Project element	Description
Candidate	S Gokul
Domain	Artificial Intelligence • Computer Vision • IoT • Physical AI
Primary objective	Occupancy-aware energy management
Potential platform	Python, OpenCV, YOLO or equivalent, ESP32 / Arduino
Physical interface	Isolated relay module with appropriate electrical protection

Problem being addressed

Classrooms and laboratories may consume electricity when they are empty or only partly occupied. Manual switching and fixed timers do not always reflect actual room usage. A system that estimates occupancy and considers usage patterns can support more informed energy decisions.

Proposed solution

- Capture the monitored space using a camera, subject to privacy rules and institutional approval.
- Detect people using a computer-vision model and convert detections into occupancy information.
- Combine current observations with historical usage, timetable and time-of-day information.
- Estimate near-future occupancy and determine an appropriate energy-control action.
- Send the decision to an edge controller, which operates a properly isolated switching interface.

04 | FEATURED PROJECT

System Workflow, Technologies and Expected Impact

The proposed system follows a perception-to-action architecture.

Step	Function
01 Sense	Camera and optional environmental sensors collect information about the room.
02 Understand	Computer vision detects people and estimates current occupancy.
03 Predict	Historical observations and timetable features help estimate near-future usage.
04 Decide	A decision layer selects an appropriate operating state for the electrical loads.
05 Act	An edge controller operates an isolated relay interface under safe control conditions.
06 Improve	Occupancy and energy data are recorded for evaluation and future refinement.

Core technologies

Layer	Possible technologies
Programming	Python
Computer vision	OpenCV; YOLO or an equivalent person-detection model
Machine learning	Scikit-learn or PyTorch
Data handling	NumPy and Pandas
Edge / IoT	ESP32 or Arduino; serial communication or MQTT
Physical actuation	Isolated relay module and protected electrical interface
Optional dashboard	Flask or React

Expected outcomes

- Real-time occupancy estimation for selected rooms.
- Reduced unnecessary operation of lights and fans where the system is safely deployed.
- A record of occupancy and energy-use patterns for evaluation.
- A scalable concept for smart classrooms and other campus spaces.
- A practical demonstration of AI perception, decision-making and physical actuation.

Safety and privacy note: Any real deployment must receive institutional approval, protect camera data, use suitable electrical isolation and be designed, installed and tested by qualified personnel. The project is a proposal, not a claim of completed deployment.

English Proficiency

CEFR results from two independent online assessments.

CEFR LEVEL

B1

Preply Online English Test

Vocabulary-based level assessment

Level: Intermediate
Words known: 2,750 at this level
To reach B2: about 1,000 more words
Meaning: can understand and use sentences on familiar or personally interesting topics from work, school or travel.

CEFR LEVEL

B1

Speak & Improve, Test 1

Research project, speaking assessment

Overall level: B1
Checked: Saturday, 3 October 2026, 10:19
Part 1 (answering questions about yourself): A2
Feedback: a good result; further speaking practice recommended.

Where I stand on the CEFR scale

A1	A2	B1	B2	C1	C2
Beginner	Elementary	Intermediate	Upper-int.	Advanced	Proficient

Current level: B1 (Intermediate)

Assessment	Date	Result	Next target
Preply online test	Not recorded	B1 (Intermediate), 2,750 words	B2: 1,000 more words
Speak & Improve, Test 1	3 Oct 2026	B1 overall; A2 on Part 1	Keep practising speaking skills

Sources: Preply online English test and the Speak & Improve research project.

Reading assessment

A Swiftread assessment measured how quickly I read and how much I retain, a useful companion to the vocabulary and speaking levels above.

READING SPEED

475 WPM

COMPREHENSION

67%

05 | ENGLISH LEVEL RECORD

Soft Skills Coursework

NPTEL soft skills lectures from IIT Kanpur.

Course	Enhancing Soft Skills and Personality (NPTEL, IIT Kanpur)
Instructor	Prof. T. Ravichandran
Format	8 weeks, roughly 40 lectures of about 30 minutes each
Covered here	Lectures 1 to 3: two recap lectures and the first lesson on mindset

> ***"Circumstances do not make a person, they reveal him."***

Lecture 1: Introduction 1

This lecture condenses the first half of the earlier course, Developing Soft Skills and Personality, so that newcomers can follow along. Learning here means a lasting change in behaviour, approached across the physical, mental, emotional, social and spiritual sides of a person.

PERCEPTIONS AND SELF-MANAGEMENT

Soft skills are skills for dealing with people, and around 80% of our happiness depends on others. From Covey: start with the end in mind and first try to understand. Next come self-awareness, confidence, emotional balance and coping with stress.

CONFLICT, STRESS AND HABITS

Look for win-win results and say what you need rather than demanding a particular fix. Stress is a mismatch between demands and resources. A habit forms in the chain thought, action, reward, habit, character, so swap bad habits for good ones.

05 | ENGLISH LEVEL RECORD

Summary of Interchange Exercises

A short reflection on language learning through classroom practice.

Describing the past

- Through the Interchange activities, I developed a clearer way of describing things that happened earlier.
- I learnt to select suitable verb forms when talking about finished events and past situations.
- The practice gave me confidence to explain where someone was and what actions took place before.
- Using familiar examples helped me understand the language through context rather than rules alone.
- I became more accurate when forming simple sentences about experiences from the past.

Routines, habits and change

- I also learnt how to describe old routines, repeated actions and situations that are no longer the same.
- The conversation activities showed me how people naturally speak about childhood memories and habits.
- I could relate the examples to my own life, which made the learning easier to remember.
- I practised showing the difference between something I did before and what I do at present.
- The exercises also helped me recognise how questions and negative sentences are formed in conversation.

Accuracy and confidence

- I became more careful about word order and choosing the right form of a verb.
- Repeated practice made me more comfortable responding to questions about earlier experiences.
- I learnt that grammar becomes easier when it is used to communicate a real memory or idea.
- This work strengthened both my sentence-building ability and my confidence while speaking English.
- Overall, the activities helped me express past experiences, habits and personal changes more naturally.

NPTEL Learning Summary | Weeks 1–4

Professional English: communication, presentation, visual awareness and interpersonal foundations.

Week	Learning themes	My learning summary
1 | Communication basics	Communicative environment; purposeful listening; appropriate speaking; starting and sustaining conversations.	I learned that effective communication begins with understanding the situation and listening carefully. Speaking with a clear purpose helps a conversation remain meaningful.
2 | Presentation and interaction	Organising content; presentation methods; multimedia basics; communication styles; group speaking.	I learned to organise information for an audience and present ideas in a logical order. Group discussions also require confidence, preparation and respect for other viewpoints.
3 | Visual and nonverbal communication	Visual culture; perception; sound; body language; facial expressions.	I understood that communication is not limited to words. Gestures, facial expressions, posture and tone can influence how a message is received.
4 | Individuals, groups and cultures	Relationship building; group dynamics; conflict and resolution; social networks and media.	I learned that teamwork depends on trust, shared responsibility and awareness of different perspectives. Conflicts can be managed through calm discussion and clear communication.

Application to engineering studies

These topics are useful when explaining a technical idea, discussing project responsibilities, presenting a prototype or asking for feedback. Clear communication helps a team understand the problem, the proposed solution and the next action.

NPTEL Learning Summary | Weeks 5–8

Developing emotional awareness, critical thinking, leadership and practical workplace habits.

Week	Learning themes	My learning summary
5 | Emotional and social skills	Understanding others; responding appropriately; positive interaction; relationship development.	I learned that emotional awareness and empathy support better communication. Respectful responses can strengthen relationships in academic and professional settings.
6 | Creativity and problem-solving	Creativity; critical thinking; analysing situations; evaluating alternatives.	I learned to examine a problem, consider different approaches and choose a solution based on reasoning rather than assumptions.
7 | Motivation, persuasion and leadership	Self-motivation; persuasion; negotiation; leadership; motivating others.	I understood that motivation supports consistent effort, persuasion requires clear reasons, negotiation involves considering different needs, and leadership includes guiding a group responsibly.
8 | Essential and vocational skills	Time management; stress management; resilience; work-life balance; workplace soft skills.	I learned that planning, realistic priorities and resilience help manage academic responsibilities. These habits are important for long-term professional growth.

Overall reflection

The eight-week learning journey helped me connect communication with everyday academic work. Technical knowledge is important, but the ability to explain ideas, listen to others, work through disagreements and manage time is equally valuable. I intend to apply these skills during project planning, team activities and future professional opportunities.

Future application

- Prepare a clear outline before giving a presentation.
- Listen fully before responding during group discussions.
- Break complex project tasks into smaller, manageable steps.
- Use feedback to improve both technical work and communication.
- Plan study and project time while allowing space for rest and recovery.

07 | COMPANY STUDY

Biographical Note on Doozy Robotics

A study of a Physical AI and humanoid robotics vision, prepared from the supplied student note.

Profile item	Details in the supplied note
Company	Doozy Robotics
Location	Singapore
Industry	Physical AI and humanoid robotics
Co-Founder and CEO	Suresh Chandrasekar
Co-Founder and COO	Ajmal Thahseen
Stated vision	Intelligent autonomous workforces

Introduction

The supplied biographical note describes Doozy Robotics as a Singapore-based company focused on Physical AI and humanoid robotics. Its stated vision is to develop intelligent autonomous workforces for sectors such as manufacturing, healthcare and logistics. The idea reflects a broader movement towards machines that can interpret their surroundings and perform useful physical tasks.

Why Physical AI matters

Physical AI connects artificial intelligence with machines that sense and act in the real world. A capable robotic system may combine computer vision, sensor fusion, motion planning, navigation, control, embedded computing and reliable software. Humanoid robots are particularly interesting because they are intended to operate in spaces designed around people, although safe and dependable operation remains a major engineering challenge.

Potential application areas

- **Manufacturing:** material handling, inspection and repetitive physical operations.
- **Healthcare:** logistical and support tasks, with safety and human supervision as essential requirements.
- **Logistics:** movement, sorting and handling of goods in operational environments.

Engineering challenges

Robots must handle changing environments, operate safely near people, respond in real time and perform tasks reliably. Hardware durability, energy use, cost, software quality and responsible deployment also influence whether a robotic system is practical.

Source note: *Company and leadership details above are reproduced as claims from the user's supplied biographical note and have not been independently verified in this portfolio.*

08 | PROFESSIONAL COMMUNICATION

Self-Introduction and Closing Reflection

A concise introduction suitable for an academic presentation or professional interaction.

Professional self-introduction

> *"Good morning. My name is S Gokul, and I am a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering. I am interested in artificial intelligence, robotics, computer vision and intelligent systems. My long-term goal is to become a Physical AI Engineer and work on systems that can sense their surroundings, make decisions and interact with the real world. I am currently strengthening my foundations in programming, problem-solving and computer science while gradually learning about robotics, embedded technologies and machine learning. I believe that consistent practice, practical projects and continuous learning will help me develop as an engineer. Thank you."*

Personal development commitments

- Build strong foundations in programming, mathematics and computer science.
- Practise explaining technical ideas in simple, organised language.
- Develop practical experience through projects, simulations and experiments.
- Accept feedback, review mistakes and improve the next version of my work.
- Work responsibly with others and consider safety, privacy and reliability in engineering.

Conclusion

This portfolio brings together my academic profile, career direction, self-assessment, project proposal, English level records, learning reflections and company study. It represents my current stage as a student and the direction in which I intend to grow. My aim is to progress step by step from programming fundamentals to the design of intelligent systems that connect AI with physical machines.

> ***"Learn the fundamentals. Build with purpose. Improve through every iteration."***

STUDENT: S GOKUL | ROLL NO.: 26CS152 | CSE - C
St. Joseph's College of Engineering • Chennai

Vocabulary Worksheet — Page 1 of 15

You Choose, We Do It

St. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution)

St. Joseph's Group of Institutions

OMR, CHENNAI - 119

VOCABULARY WORK SHEET

Part A: Fill in the Blanks

Complete the sentences using the correct terms from Units 1, 2, and 3.

1. To find out what someone's job is, you usually ask, 'What do you do?'
2. A manager who oversees a team and ensures targets are met is often said to be in charge of the department.
3. Employees who work outside standard office hours in rotation (e.g., one week morning, next week evening) are working in shifts.
4. Outside specialists hired by companies to persuade top executives to leave their current firms are known as headhunters.
5. Job applicants typically submit a CV (curriculum vitae) detailing the story of their working life alongside a covering letter.

Part B: Matching Terms to Definitions

Match each business term on the left with its correct description on the right.

1. B_____ Flexitime
2. C_____ Teleworking
3. E_____ Psychometric test
4. D_____ Shortlist
5. A_____ Referees

- **A.** Previous employers or teachers named to provide character or performance references.
- **B.** A system allowing employees to choose working hours within agreed limits.
- **C.** Working from home using a computer and telecommunications.
- **D.** A selected pool of qualified candidates chosen for final interviews.
- **E.** A written test used to assess a candidate's intelligence and personality.

Part C: Sentence Correction

Identify and correct the errors in the following sentences.

1. *Error:* "I am in charge for the design department." → *Correction:* I am in charge of the design department.

Vocabulary Worksheet — Page 2 of 15

2. *Error:* "He is going to the work by train every morning." → *Correction:*
   He goes to work by train every morning.
3. *Error:* "She is a very responsible in her daily tasks." → *Correction:*
   She is very responsible in her daily tasks.

Part A: Fill in the Blanks

Complete the sentences using the correct terms from Units 4 and 5.

1. Someone who has just left university with a degree is referred to as a graduate.
2. A worker who requires virtually no specialized training or formal qualifications is classified as unskilled.
3. Salespeople often receive a basic salary plus commission, which is a percentage on everything they sell.
4. Senior executives may receive share (stock) options, giving them the right to buy company shares at low prices.
5. High-ranking corporate executives with excessive compensation packages are sometimes referred to informally as fat cats.

Part B: Matching Collocations

Match the adjectives/phrases on the left with their correct prepositional complements.

1. E_____ Skilled _____
2. D_____ Good _____
3. A_____ Computer _____
4. B_____ Self _____
5. C_____ Fringe _____

- **A.** literate
- **B.** starter / motivated
- **C.** benefits
- **D.** at / in
- **E.** with

Part C: Application & Context Analysis

Explain the difference between the following pairs in 1-2 sentences:

1. **Salary vs. Wages:**
   A salary is normally a fixed regular payment, often monthly; wages are usually calculated by hours or days worked.
   _____
2. **Severance package vs. Bonus:**
   A severance package is compensation given when employment ends; a bonus is extra pay for performance or results.
   _____

Part A: Fill in the Blanks

Vocabulary Worksheet — Page 3 of 15

Complete the sentences using terms from Units 6, 7, and 8.

1. Manual workers who use their hands and wear blue collars are often contrasted with white-collar workers in offices.
2. When workers stop working temporarily to protest pay or conditions, they go on strike.
3. A company that reduces its management layers and makes itself leaner is said to have delayered.
4. When an employee is dismissed through no fault of their own because their job no longer exists, they are made redundant.
5. An invisible barrier that prevents women from rising beyond a certain corporate level is called the _ glass __ ceiling.

Part B: Matching Health & Safety / Legal Terms

Match each workplace issue with its accurate classification.

1. B_____ Repetitive Strain Injury (RSI)
2. D_____ Passive smoking
3. A_____ Sexual harassment
4. C_____ Equal opportunities
5. E_____ Whistleblowing

- **A.** Unwelcome sexual behavior in the workplace
- **B.** Physical injury caused by continuous data entry or typing
- **C.** British term equivalent to US affirmative action
- **D.** Breathing in other people's tobacco smoke involuntarily
- **E.** Reporting illegal or unethical practices within an organization

Part C: Short Answer Exercises

1. What is the difference between being **dismissed** and being **made redundant**?
   Dismissal means an employer ends someone's employment, often because of conduct or performance. Redundancy happens when the job itself is no longer needed.
   _____
2. What is an **open-plan office**?
   An open-plan office is a shared workspace with few or no dividing walls between desks.
   _____

Part A: Fill in the Blanks

Complete the sentences using terms from Units 9, 10, and 11.

1. Directors who are outsiders and do not participate in the day-to-day running of the company are non-executive directors.
2. In the US, the top manager is often the CEO, while chief operating officers (COOs) take care of the day-to-day running.

3. Someone who founds or establishes their own company and takes financial risks is an entrepreneur.
4. Powerful leaders who control massive business empires are often called magnates, moguls, or tycoons.
5. Businesses that are sold off by the state to private investors are privatized.

Part B: Matching Corporate Terms

Match the terms on the left with their definitions.

1. A_____ SME
2. D_____ Corporate culture
3. E_____ Enterprise zone
4. B_____ Head up
5. C_____ Boardroom

- **A.** Small and Medium-sized Enterprise
- **B.** To head or lead a department or activity
- **C.** The room where directors meet
- **D.** The shared beliefs and working style of a company
- **E.** A designated region with fewer laws and lower taxes to encourage business

Part C: Error Correction

Correct the italicized phrases:

1. "He is the head up of our marketing department." →
   He heads up our marketing department.
2. "We need to buy a commerce for our new online shop." →
   We need to set up an e-commerce business for our new online shop.

Part A: Fill in the Blanks

Complete the sentences using terms from Units 12, 13, and 14.

1. A self-employed person working alone, especially a builder or plumber, is a sole trader.
2. A company whose shares can be freely bought and sold on the stock market is a Public Limited Company or PLC.
3. Organizations with social aims like helping the sick or poor are non-profit organizations or charities.
4. Software developers often release a final test version, called a beta version, before official launch.
5. If a design defect is discovered after a product launch, a company may have to recall it.

Part B: Sector Classification

Vocabulary Worksheet — Page 5 of 15

*Classify each industry as either **Manufacturing (M)** or **Services (S)**.*

1. 1·M Aerospace
2. 2·S Catering
3. 3·M Pharmaceuticals
4. 4·S Property / Real Estate
5. 5·M Steel production

Part C: Short Definitions

1. Define **CADCAM**:
   CADCAM means Computer-Aided Design and Computer-Aided Manufacturing; computers help design and produce goods.
   _____
2. Define **Focus group**:
   A focus group is a small group of people asked to discuss a product, idea, or advertisement to provide feedback.
   _____

Part A: Fill in the Blanks

Complete the sentences using terms from Units 15, 16, and 17.

1. Products that are no longer up-to-date because they use old technology are obsolete.
2. Legal protection for an invention that stops others from copying its design is a patent.
3. The maximum amount that a plant or company can produce is its capacity.
4. Materials, components, labor, and capital used in production are collectively known as inputs.
5. Manufacturing goods efficiently without waste by having components arrive exactly when needed is just-in-time (JIT) production.

Part B: Matching Terms

1. B_____ R&D
2. A_____ Royalties
3. C_____ Craft industry
4. E_____ Work-in-progress
5. D_____ Outsourcing

- **A.** Payments made to an author or inventor for using their protected work
- **B.** Research and Development department
- **C.** Manual, labor-intensive production of items like furniture
- **D.** Using outside suppliers for services previously done in-house
- **E.** Goods that are currently being made on the production line

Part C: Short Explanations

Vocabulary Worksheet — Page 6 of 15

1. Why do companies implement **Just-in-Time (JIT)** manufacturing?
   Companies use JIT to reduce inventory and storage costs, minimize waste, and receive materials only when they are needed.

Part A: Fill in the Blanks

Complete the sentences using terms from Units 18, 19, and 20.

1. The objective of quality control is conformance to specifications with zero defects.
2. Comparing your performance against the best operators in the industry is called benchmarking.
3. People who buy everyday services like train travel are customers, while those who buy professional services are clients.
4. A company that sells more than any other competitor in a market is the market leader.
5. When competition is extremely strong and aggressive, it is often described as cut-throat or fierce.

Part B: Matching Terms

1. B_____ Kaizen
2. C_____ Mystery shopper
3. E_____ Market segment
4. D_____ Key players
5. A_____ BPR

- **A.** Business Process Re-engineering
- **B.** Japanese term for continuous improvement
- **C.** Person pretending to be a customer to check service quality
- **D.** The most important companies in a particular market
- **E.** A distinct group of buyers targeted within a market

Part C: Analysis Task

1. What is the difference between a **market leader** and a **challenger**?
   A market leader has the largest market share; a challenger is a competitor trying to increase its share and overtake the leader.

Part A: Fill in the Blanks

1. The four Ps of marketing are product, price, place, and promotion.
2. Consumer goods that last a long time, such as cars and washing machines, are consumer durables.
3. Products that do not have a brand name are known as unbranded products or generics.
4. Cheap items sold at or below cost to attract customers into a store are loss leaders.

5.· A small group of buyers with specialized needs forms a niche market.

Part B: Matching Terms

1. A_____ FMCG
2. B_____ Own-brand
3. D_____ Upmarket
4. C_____ Trade up
5. E_____ Price war

- **A.** Fast-Moving Consumer Goods (fast-selling items like food)
- **B.** Products sold under the retailer's own name
- **C.** Moving from basic models to more sophisticated, expensive ones
- **D.** Designed for higher-income users; upscale
- **E.** When competing companies continually reduce prices in response to each other

Part C: Short Answer

1. Name the **Four Ps** of the marketing mix:
   Product, Price, Place and Promotion.

_____

Part A: Fill in the Blanks

Complete the sentences using terms from Units 24, 25, and 26.

1.· Wholesalers and retailers are collectively known as intermediaries.
2.· A retail store that is part of a group of shops all sharing the same name is a chain store.
3. Unsolicited direct mail sent to consumers' homes is often dismissively called junk mail.
4. A series of advertisements for a particular product or company is an advertising campaign.
5. Businesses combining traditional physical shops with online e-commerce operations practice bricks-and-mortar retail.

Part B: Matching Terms

1. C_____ Cold call
2. E_____ Franchise
3. A_____ Billboard (AmE)
4. D_____ Loyalty card
5. B_____ B2B

- **A.** Open-air hoarding used for advertising
- **B.** Business-to-Business e-procurement
- **C.** Unannounced sales phone call to someone with no prior contact

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- **D.** Card offering reward points based on customer spending
- **E.** Business format run by a franchisee under a franchisor's brand

Part A: Fill in the Blanks

1. Money received from sales is known as revenue or turnover.
2. Costs that do not change when production goes up or down (e.g., rent) are fixed costs.
3. Products that have very good profitability and generate large cash surpluses are cash cows.
4. Customers who owe money to a business are recorded as accounts receivable.
5. Independent outside accountants who examine a company's accounts to see if they give a true and fair view are auditors.

Part B: Matching Financial Terms

1. A_____ COGS
2. C_____ Goodwill
3. B_____ Depreciation
4. E_____ In the red
5. D_____ Window dressing

- **A.** Cost of Goods Sold (variable manufacturing costs)
- **B.** Reduction in the book value of fixed assets over time
- **C.** An intangible asset representing a company's good reputation
- **D.** Creative accounting making results look better than they are
- **E.** Operating at a financial loss or having a negative bank balance

Part A: Fill in the Blanks

1. The person who is in charge of running a meeting and keeping participants on track is the chairperson or chair, often referred to simply as the chair.

2. A list of items to be discussed during a meeting, sent out in advance, is called the agenda.

3. When negotiators or meeting participants reach a middle-ground solution where both sides make concessions, they reach a compromise.

4. Using cautious language to avoid making overly direct or absolute statements is known in discussion techniques as hedging.

5. When a meeting discussion completely breaks down and no agreement can be reached, it is referred to as a deadlock.

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Part B: Matching Meeting Terms

1. _C_____ AGM
2. _E_____ Minutes
3. _A_____ Consensus
4. _B_____ Action points
5. _D_____ Hedging

- **A.** General agreement reached by all participants
- **B.** Specific tasks assigned to individuals to complete after a meeting finishes
- **C.** Annual General Meeting held once a year for shareholders
- **D.** Avoiding direct or committal statements using qualifying language
- **E.** The written record of decisions and discussions made during a meeting

Part C: Short Answer Exercises

1. What is the main responsibility of a meeting chairperson before a meeting begins?
The chairperson should plan the meeting, prepare an agenda and ensure participants receive the relevant information in advance.

2. What is the difference between reaching a **consensus** and voting on a decision?
Consensus means the group reaches general agreement through discussion; voting decides an issue by counting individual choices.

Part A: Fill in the Blanks

1. The visual items used during a presentation, such as slides, handouts, or overhead projections, are called visual aids.

2. Establishing a positive connection and good mutual relationship with your audience during a presentation is building rapport.

3. The opening part of a presentation where you welcome the audience and outline your structure is the introduction.

4. Moving around too much or making nervous hand gestures can distract from your body language during a speech.

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5. When concluding a presentation, speakers typically open the floor to deal with questions from the audience.

Part B: Matching Presentation Stages

1. _C_____ Visual aids
2. _B_____ Body language
3. _D_____ Rapport
4. _E_____ Q&A session
5. _A_____ Handouts

- **A.** Printed materials distributed to listeners to supplement a presentation
- **B.** Postures, gestures, and facial expressions used while speaking
- **C.** Charts, slides, and diagrams used to illustrate key points visually
- **D.** A harmonious, trusting relationship between speaker and listeners
- **E.** The time at the end of a presentation dedicated to answering inquiries

Part C: Error Correction

Correct the italicized phrases:

1. "During my presentation, I will show you some visual supports." → During my presentation, I will show you some visual aids.

2. "Let's open the floor for some inquiries from the spectators." → Let's open the floor for questions from the audience.

Part A: Fill in the Blanks

Complete the sentences using terms from Units 62 through 66.

1. A negotiation style where both sides aim for an outcome that benefits everyone involved is known as a win-win situation.

2. Offering something you value less in exchange for something the other party values less during talks is making trade-offs.

3. When negotiations completely stall and neither side is willing to move or compromise, they have reached a deadlock.

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4. An agreement between two or more parties that is enforceable by law after a successful negotiation is a contract.

5. A neutral third party brought in to help resolve a deadlock and mediate an agreement is a mediator.

Part B: Matching Negotiation Tactics

1. _A_____ Probing

2. _D_____ Counter-proposal

3. _B_____ Deadlock

4. _E_____ Concession

5. _C_____ Mediator

- **A.** Asking probing questions to discover the other party's underlying needs
- **B.** A stall in talks where neither side will compromise
- **C.** An independent third party assisting in dispute resolution
- **DT.** A responding offer made in reaction to an initial proposal
- **E.** Something given up voluntarily to reach an agreement

Part C: Short Answer Exercises

1. What is the difference between a **hard bargaining** style and a **win-win** style?
Hard bargaining focuses on winning as much as possible for one side; a win-win style seeks a mutually beneficial agreement.
_____

2. Why is checking the final contract details essential after reaching an agreement?
Checking contract details ensures that all terms are accurate, understood, lawful, and consistent with what was agreed.
_____

Part A: Fill in the Blanks

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Complete the sentences using terms from Units 36 and 37.

1. When a company sells its shares to the public for the first time, it floats the company in a flotation / initial public offering (IPO).

2. Short-term lending to businesses is known as commercial credit.

3. Financial products derived from underlying assets, such as futures and options contracts, are known as derivatives.

4. An extended period of rising share prices and optimistic dealer sentiment is referred to as a bull market.

5. Shares in long-established, stable companies with a history of profit in both good and bad times are called blue chips.

Part B: Matching Financial Markets

1. _B_____ Forex

2. _C_____ Bourse

3. _A_____ NASDAQ

4. _E_____ FTSE

5. _D_____ Bear market

- **A.** New York stock market index focusing heavily on high-tech companies
- **B.** Foreign exchange market for buying and selling national currencies
- **C.** An alternative term for a stock market or stock exchange
- **D.** A market characterized by falling prices and pessimistic dealer sentiment
- **E.** London's Financial Times Stock Exchange index

Part C: Short Answer Exercises

1. What is the difference between a **futures contract** and an **options contract**?
A futures contract generally obliges the parties to transact later at agreed terms; an option gives its buyer the right, but not the obligation, to transact.

2. Why are major financial centres like London and New York often referred to by nicknames like "The City" or "Wall Street"?

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The City and Wall Street are famous financial districts, so their names are used as shorthand for the financial industries based there.

Part A: Fill in the Blanks

1. The study of how money works, is used, and whether economic activities are profitable is called economics.

2. When a country exports more goods and services than it imports, it maintains a trade surplus.

3. The total monetary value of all goods and services produced within a particular country is its Gross Domestic Product (GDP).

4. A period of negative economic growth where economic output contracts is known as a recession.

5. When a bad recession becomes extremely severe, widespread, and prolonged, it is called a slump or depression.

Part B: Matching Economic Indicators

1. _C_ inflation rate ion rate
2. _D_ Trade deficit
3. _E_ Stagflation
4. _B_ Trough
5. _A_ Stagnation

- **A.** A period when the economy is growing very slowly or not at all
- **B.** The lowest point in an economic cycle before output starts rising again
- **C.** The speed at which general consumer price levels are rising
- **D.** When import payments exceed export earnings, creating a trade gap
- **E.** A difficult economic condition combining slow growth with rapid inflation

Part C: Error Correction

Vocabulary Worksheet — Page 14 of 15

Correct the italicized phrases:

1. "The country's economic indicators show a high rate of inflation speeds." → The country's economic indicators show a high rate of inflation.

2. "When output reaches its lowest level, it hits a peak before rising." → When output reaches its lowest level, it hits a trough before rising.

Part A: Fill in the Blanks

1. Buying or selling securities using confidential information that is not publicly available is illegal insider dealing.

2. A secret agreement among competing companies in the same market to fix prices at a certain level is price fixing.

3. A type of fraud where an employee illegally steals and gets money from their employer is known as embezzlement.

4. The process of hiding the illegal origin of criminal money, often by purchasing property, is money laundering.

5. Directing investment funds exclusively into companies that do not harm people or the environment is ethical investment.

Part B: Matching Crime & Ethics Terms

1. D Embezzlement
2. A Counterfeiting
3. B Bribery
4. E Code of ethics
5. C Sweatshop labor

- **A.** Making illegal fake copies of banknotes, documents, or luxury goods
- **B.** Offering or receiving illegal payments to persuade someone to act corruptly
- **C.** Manual labor performed under severely exploitative, underpaid conditions
- **D.** The illegal theft of funds by someone entrusted with managing them within a firm
- **E.** A formal corporate document outlining required moral standards of behavior

Vocabulary Worksheet — Page 15 of 15

Part C: Short Answer Exercises

1. What is the purpose of establishing **Chinese walls** within financial institutions?
Chinese walls separate departments and restrict confidential information to prevent conflicts of interest and insider dealing.

2. Why do institutional investors conduct annual **social performance audits**?
Social performance audits check whether companies meet ethical, social and environmental standards and help investors assess risks.

Part A: Fill in the Blanks

1. The overall period during which a project or task should be completed is referred to as the timescale or timeframe.

2. When work is completed later than the planned completion date, it is behind schedule and there is a delay.

3. Work that is difficult in an interesting, rewarding, and enjoyable way is described as challenging.

4. An extreme state of physical and mental exhaustion caused by chronic work stress is known as burnout.

5. The metaphor used to describe the exhausting, highly competitive modern corporate work environment is the rat race.

Part B: Matching Time & Stress Terms

1. D Downtime
2. E Prioritize
3. C Perfectionism
4. A Downshifting
5. B Stress-induced

- **A.** Choosing to leave behind a stressful career for a simpler, balanced lifestyle
- **B.** An unexpected illness or condition brought on by excessive pressure
- **C.** An unworkable aim for flawlessness when there is no need for it

10 | REFLECTION

Vocabulary Skills in My Workspace

How the vocabulary practised in this worksheet will make me a stronger professional.

Working through this vocabulary worksheet showed me that the right word does real work in a professional setting. Knowing terms from areas such as meetings, presentations, negotiation, finance, marketing and workplace ethics helps me understand what colleagues, managers and clients actually mean, so I can follow discussions without hesitation and ask sharper questions. In my own work, a strong vocabulary will help me write clear emails and project reports, explain a technical idea in simple language, present a prototype with confidence, and take part in team discussions by contributing precise points instead of vague ones. It will also help me handle sensitive moments, such as disagreeing politely, using careful language when I am unsure, and reaching a compromise in a negotiation, because I will have the exact phrases to do so. As I grow towards a career in Physical AI and robotics, where engineers must explain complex systems to teammates, managers and non-technical stakeholders, these skills will help me earn trust, avoid misunderstandings and communicate my ideas with purpose. I intend to keep building this habit by noting new terms each week and using them in my daily writing and speaking, so that my communication improves along with my technical ability.

Interchange Digital Workbook Progress

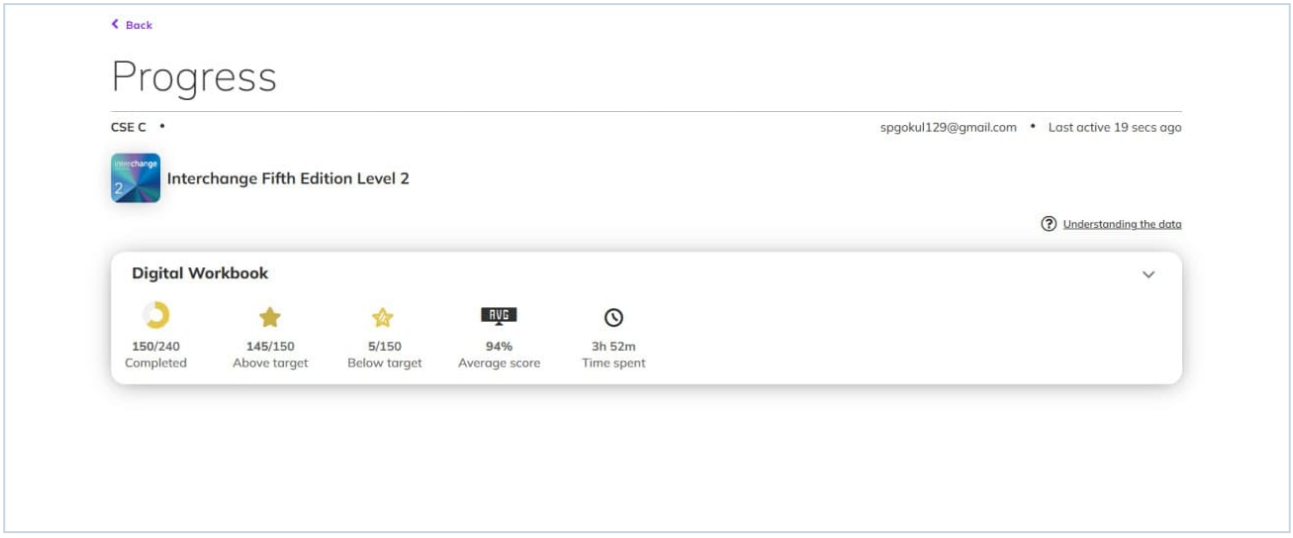
Interchange Fifth Edition Level 2 | Class: CSE C

COMPLETED	ABOVE TARGET	BELOW TARGET	AVERAGE SCORE	TIME SPENT
150/240	145/150	5/150	94%	3h 52m

Progress summary

In the Digital Workbook for Interchange Fifth Edition Level 2, I have completed 150 of 240 activities. Of these, 145 were scored above target and only 5 below target, giving an average score of 94% over 3 hours 52 minutes of practice. This steady, accurate work supports the speaking and grammar gains described in my Interchange reflection earlier in this portfolio, and I plan to complete the remaining activities.

Platform record



Progress screen from the Interchange Digital Workbook.